



Technical Service Bulletin

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Articulated Piston Power Cylinder Update

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Warranty Statement

The information in this document has no effect on present warranty coverage or repair practices, nor does it authorize TRP or Campaign actions.

Contents

This Technical Service Bulletin introduces a new hardened cylinder liner and top piston ring for the QSK45 and QSK60 engines with articulated pistons. The new cylinder liner and top piston ring (top ring only) change is being implemented to improve power cylinder durability.

The new cylinder liner, Part Number 4085917, has been nitride hardened for improved robustness and supersedes the old cylinder liner, Part Number 3410342. The new cylinder liner is dimensionally the same as its predecessor. Use Table 1 for component part numbers and Table 2 for Service Kit part numbers.

The new top piston ring, Part Number 3410861, is a cast iron ring with a wrap around hard particle chrome. The old top piston ring, Part Number 3070589, is **not** to be used with the new hardened cylinder liner. The old piston ring is **not** obsolete because it is used on some K38, K50 and QSK19 engines. The middle and bottom piston rings remain the same and are not affected by this product improvement. Use Table 1 for component part numbers and Table 2 for Service Kit part numbers.

When the new-hardened cylinder liner is unused it is duller in appearance when compared to the old cylinder liner because of the nitride hardening. However, the only certain method to tell the difference between the cylinder liners is by the part number identification, stamped into the cylinder liner. See Fig. 1. Visual inspection of the top piston ring part number is also required to determine correct part number piston ring. Use Fig. 2.

Table 1 (Component Part Numbers)

Component Description	Old Part Number	New Part Number
Cylinder Liner	3410342	4085917
Top Piston Ring	3070589	3410861

Table 2 (Service Kit Part Numbers)		
Service Kit Description	Old Part Number	New Part Number
Cylinder Liner Kit	4024811	4089143
Piston Ring Kit	3631248	4089145

Table 3 (Service Kit Compatibility Matrix for Articulated Pistons)		
Service Kit Description Compatibility Matrix	Old Cylinder Liner Kit Part Number 4024811	New Cylinder Liner Kit Part Number 4089143
Old Piston Ring Kit Part Number 3631248	YES	NO
New Piston Ring Kit Part Number 4089145	YES	YES

Note : The old top piston ring can not be used with the new hardened cylinder liner. If this combination is used, top piston ring durability will be negatively effected.

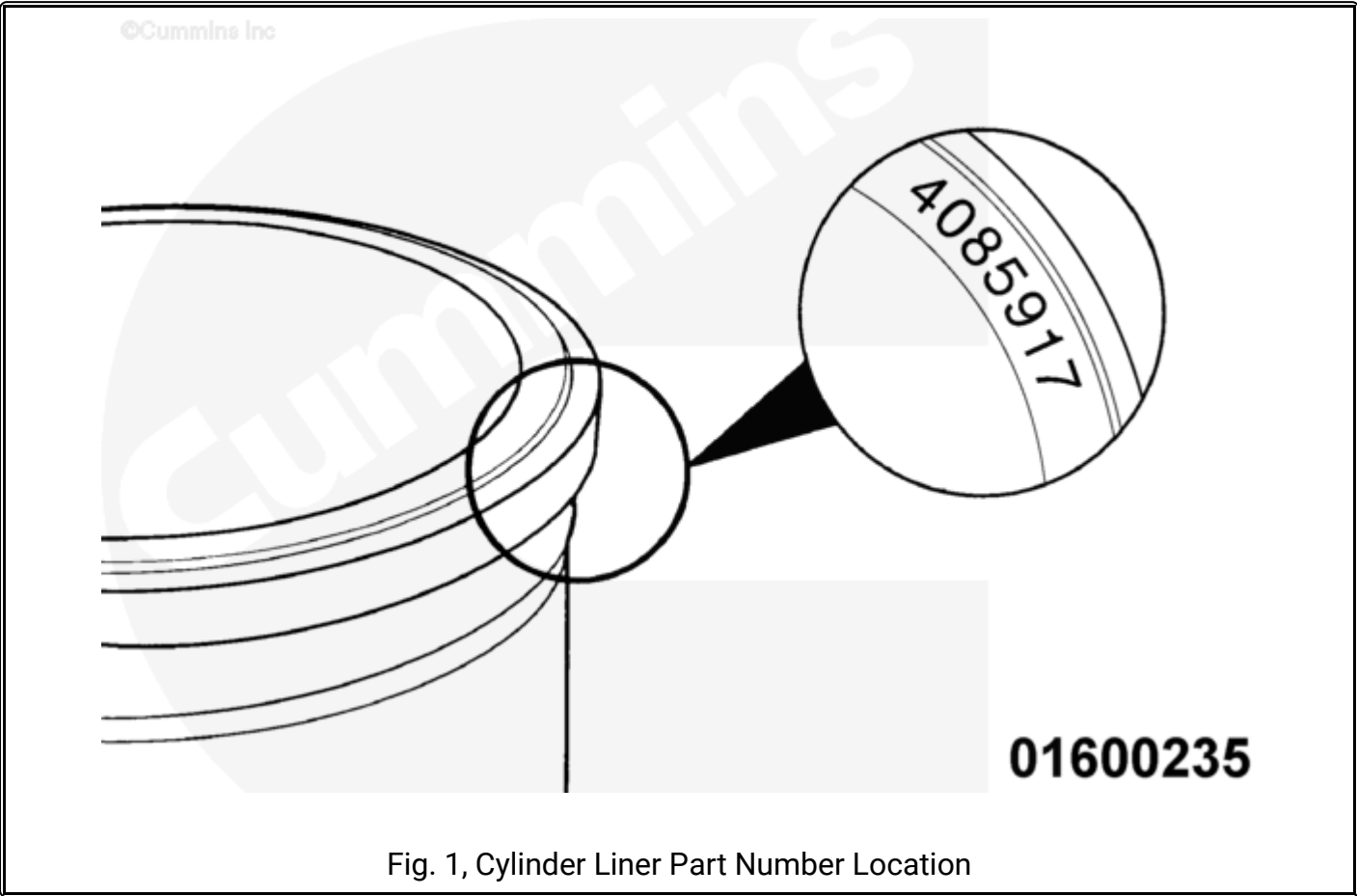


Fig. 1, Cylinder Liner Part Number Location

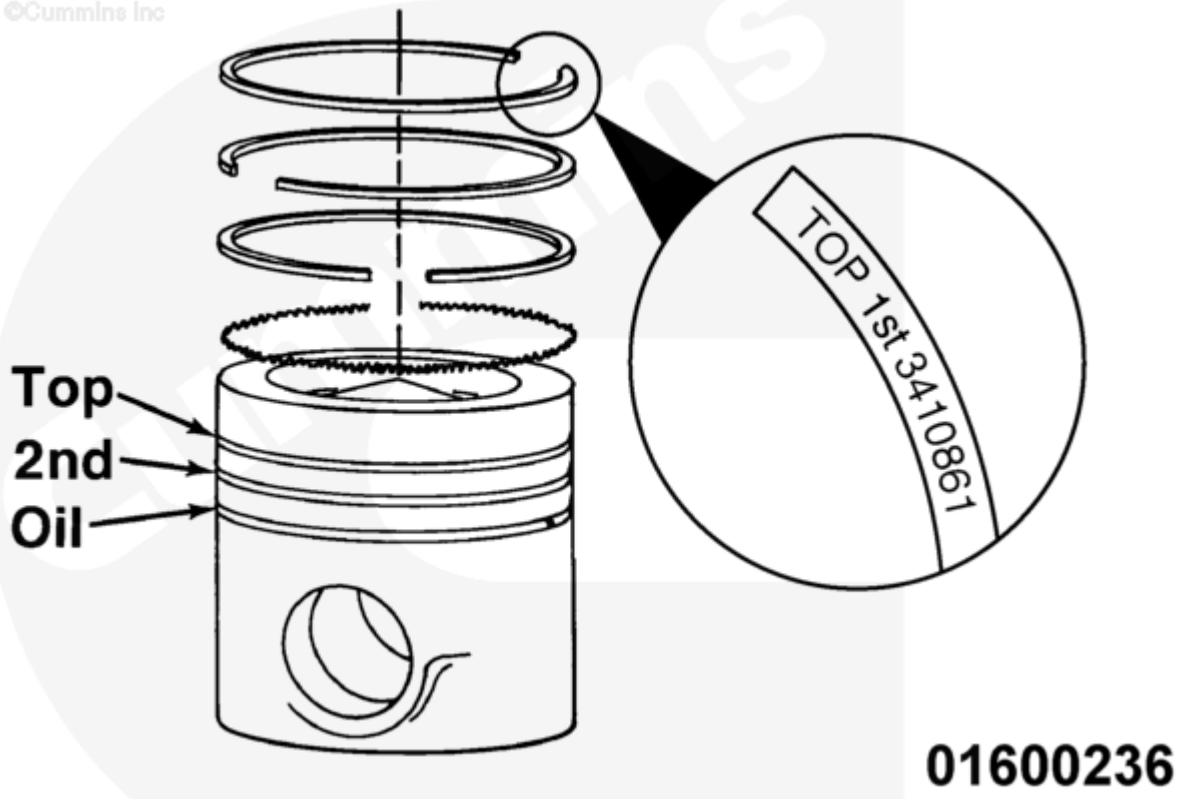


Fig. 2, Piston Ring Part Number Location

This product improvement went into production at the Daventry Engine Plant on QSK45/60 industrial engines, effective January 14, 2002. Power Generation applications are pending, but **not** available at the time of the release of this service parts topic.

Note : This is a product improvement **not** subject to campaign or temporary repair practice.